TUTORING CENTRE	Life Sciences · Physics Sheet **Mechanical Oscillations** Grade 12 LS · Chapter 4 · Student toolkit	Lebanese curriculum

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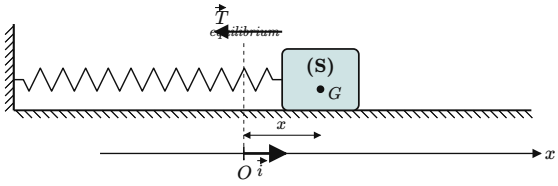
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Session 1 — The free undamped oscillator ≈ 2 h 30

1 Oscillations: the words

An **oscillation** is a to-and-fro motion about a stable equilibrium position. **Amplitude** X_m (m); **period** T (s); **frequency** $f = 1/T$ (Hz).

Free oscillations: no energy is received after the release. **Undamped**: no friction, constant amplitude. **Damped**: friction, decreasing amplitude.



2 The horizontal elastic pendulum

Block (S) of mass m, spring of stiffness k, no friction. $x = \overline{OG}$ (O: equilibrium), $v = x'$, $a = x''$. Tension $\vec{T} = -kx\,\vec{i}$ (restoring force). Reference of GPE: horizontal plane through G.

3 The differential equation — two roads

Energy. $ME = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$ is conserved (no friction): $\frac{d(ME)}{dt} = mvx'' + kxv = 0$; v not always zero $\Rightarrow x'' + \frac{k}{m}x = 0$.

Newton. $m\vec{g} + \vec{N} + \vec{T} = m\vec{a}$, with $m\vec{g} + \vec{N} = \vec{0}$: $-kx = mx'' \Rightarrow x'' + \frac{k}{m}x = 0$.

Result $x'' + \omega_0^2 x = 0$, $\omega_0 = \sqrt{k/m}$ (rad/s): simple harmonic motion.

4 The solution

Time equation $x = X_m \sin(\omega_0 t + \varphi)$, $X_m > 0$; $T_0 = \dfrac{2\pi}{\omega_0} = 2\pi\sqrt{\dfrac{m}{k}}$; $f_0 = \dfrac{1}{T_0}$. Check: $x'' = -\omega_0^2 x$.

Reading $x(t)$: X_m = highest value; T_0 = time between two successive maxima; $x(0)$ and the slope at $t = 0$ give φ.

5 Finding X_m and φ

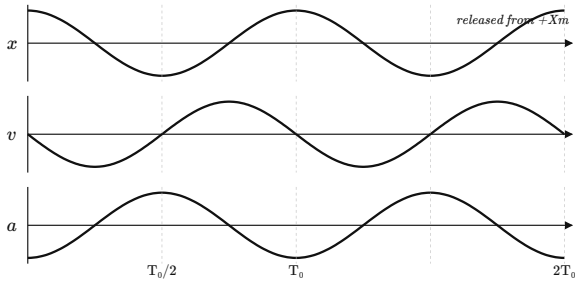
At $t = 0$: $x_0 = X_m \sin\varphi$, $v_0 = X_m\omega_0 \cos\varphi$, so $X_m = \sqrt{x_0^2 + (v_0/\omega_0)^2}$. The sign of v_0 is the sign of $\cos\varphi$.

Example. $\omega_0 = 10$ rad/s, $x_0 = 2$ cm, $v_0 = -0.35$ m/s: $X_m \approx 4$ cm, $\sin\varphi = 0.5$, $\cos\varphi < 0 \Rightarrow \varphi = \frac{5\pi}{6}$.

At $t = 0$	φ
at rest at $+X_m$	$\pi/2$
at rest at $-X_m$	$-\pi/2$
at O, + sense	0
at O, − sense	π

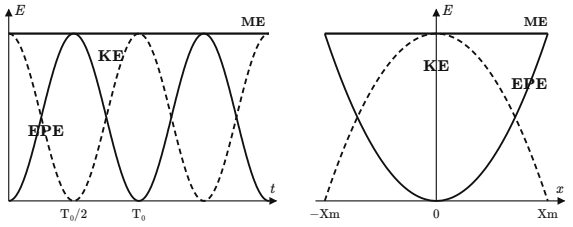
6 Velocity and acceleration

Derive $v = X_m\omega_0 \cos(\omega_0 t + \varphi)$, $V_m = X_m\omega_0$ (at O).
$a = -\omega_0^2 x$, $a_m = X_m\omega_0^2$ (at $\pm X_m$); a is always directed towards O.
$a(x)$: straight line of slope $-\omega_0^2$; $v(x)$: an ellipse.



7 Energy

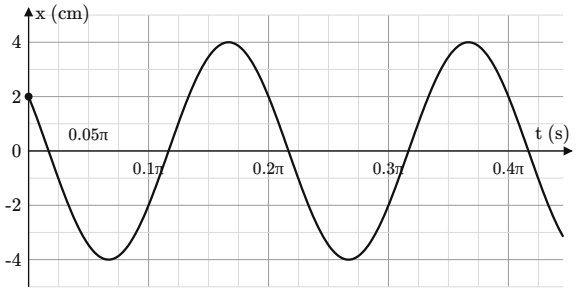
Energies $KE = \frac{1}{2}mv^2$, $EPE = \frac{1}{2}kx^2$, $ME = \frac{1}{2}kX_m^2 = \frac{1}{2}mV_m^2$ = constant. KE and EPE have the period $T_0/2$; $KE = EPE$ at $x = \pm X_m/\sqrt{2}$.



Practice 1 — after Session 1 ≈ 1 hour

Exercise 1 — A glider recorded by a motion sensor 7 pts

A glider (S) of mass $m = 250$ g, fixed to a horizontal spring of stiffness k, oscillates without friction on an air track. Doc. 1 gives the abscissa x of its centre of mass G (origin O at equilibrium).



Doc. 1

1 Specify the type of the oscillations of (S). Justify.

2 Using Doc. 1, give the amplitude X_m and the proper period T_0. Deduce ω_0 and k.

3 The time equation is $x = X_m \sin(\omega_0 t + \varphi)$. Using the value and the slope of the curve at $t = 0$, determine φ.

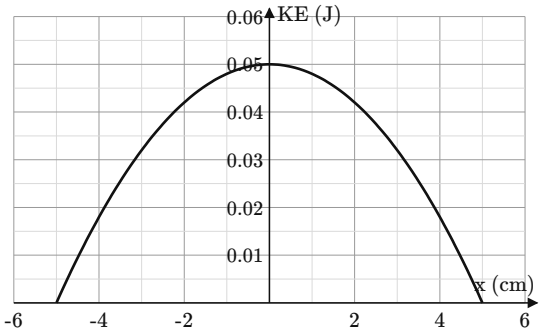
4 Determine the expression of $v(t)$, the maximum speed V_m and the initial velocity v_0.

5 Calculate the mechanical energy of the system [(S), spring, Earth]. Deduce the speed of (S) when $x = -2$ cm.

6 Calculate the acceleration of (S) when it is at $x = -X_m$. In which sense is it directed?

Exercise 2 — Energy graph of an oscillator — 7 pts

A block (S) of mass $m = 0.4$ kg, fixed to a spring of stiffness k, slides without friction on a horizontal table. At $t_0 = 0$, (S) is released without initial velocity from $x_0 = -X_m$. Doc. 2 gives the kinetic energy of (S) as a function of its abscissa x. The horizontal plane through G is the reference for gravitational potential energy.



Doc. 2

1 Write the expression of the mechanical energy of the system [(S), spring, Earth] in terms of m, k, x and v. Deduce the differential equation of the motion.

2 Using Doc. 2, give X_m and the value of the mechanical energy. Justify.

3 Deduce k, the proper angular frequency ω_0 and the maximum speed V_m.

4 Determine the abscissas where $KE = EPE$, and the speed of (S) at these positions.

5 Determine φ and write the time equation $x(t)$.

6 Sketch, on the same axes, $EPE(x)$ and $ME(x)$ for $-X_m \le x \le X_m$.

Exercise 3 — Which spring for the lab? — 6 pts

Students fix blocks of different masses m to the same spring, launch them on a horizontal frictionless track and time 10 oscillations each time.

m (kg)	0.1	0.2	0.3	0.4	0.5
$10\,T_0$ (s)	3.16	4.47	5.48	6.32	7.07

1 Why do the students time 10 oscillations instead of one?

2 Calculate T_0^2 for each mass and trace $T_0^2 = f(m)$ (1 cm $\leftrightarrow$ 0.1 kg; 1 cm $\leftrightarrow$ 0.1 s^2).

3 Show that the curve agrees with $T_0 = 2\pi\sqrt{m/k}$. Determine k.

4 An unknown block oscillates with $T_0 = 0.60$ s on the same spring. Find its mass.

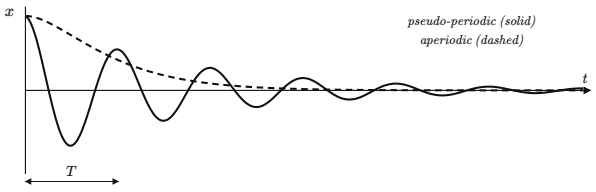
5 Without changing the spring, how should the mass change to double the frequency?

Session 2 — Real oscillators ≈ 2 h 30

8 Free damped oscillations

Friction does negative work: ME decreases ($\rightarrow$ internal energy) and the amplitude decreases. **Pseudo-periodic** (weak friction): pseudo-period $T \gtrsim T_0$, $T \approx T_0$. **Aperiodic** (strong friction): back to O without oscillating (critical = fastest).

Between two extremes ($v = 0$): $\Delta ME = \frac{1}{2}k(X_2^2 - X_1^2) < 0$.



Viscous friction $\vec{f} = -h\vec{v}$. $\frac{d(ME)}{dt} = -hv^2 \Rightarrow x'' + \frac{h}{m}x' + \frac{k}{m}x = 0$.

9 A car on its suspension

Spring + shock absorber at each wheel. A worn shock absorber (weak damping) lets the car bounce at $T_0 = 2\pi\sqrt{M/k}$; a damping close to **critical** brings the body back fastest without bouncing.

10 Driving (maintaining) the oscillations

Definition Supplying the oscillator with the energy it loses by friction so that its amplitude stays constant (pendulum clock, swing, quartz watch). Average power: $P_{av} = \dfrac{E}{\Delta t}$.

LS 2009. $k = 80$ N/m, amplitude 12.5 cm $\rightarrow$ 12 cm in 10 oscillations, $T \approx T_0 = 0.1\pi$ s: $\Delta ME = -0.049$ J, $P_{av} = \frac{0.049}{\pi} \approx 0.016$ W.

11 Measuring k or m

$T_0^2 = \dfrac{4\pi^2}{k}\,m$: $T_0^2 = f(m)$ is a straight line through O of slope $a = \frac{4\pi^2}{k}$. Astronauts "weigh" themselves this way in orbit.

12 Simple pendulum

Small oscillations: $T_0 = 2\pi\sqrt{\ell/g}$, independent of the mass. $T_0^2 = \frac{4\pi^2}{g}\ell$: slope $a \Rightarrow g = 4\pi^2/a$.

13 Exam method: collision, then oscillation

Two stages ① Shock: linear momentum conserved, $mv = (M+m)V_0$ (KE not conserved if inelastic). ② Oscillation from O: $\omega_0 = \sqrt{k/(M+m)}$, $X_m = V_0/\omega_0$, $\varphi = 0$ or π.

Practice 2 — after Session 2 ≈ 1 hour

Exercise 1 — Damped, then maintained 7 pts

A horizontal elastic pendulum: block (S) of mass $m = 0.5$ kg, spring of stiffness $k = 50$ N/m. (S) is shifted by 10 cm from O and released without initial velocity at $t_0 = 0$. After 10 pseudo-periodic oscillations the amplitude is 8 cm.

1. The oscillations are said to be free and damped. Explain both words.
2. Calculate T_0. The pseudo-period T is very close to T_0. Why?
3. Calculate the variation of the mechanical energy during these 10 oscillations. In what form does the lost energy appear?
4. The friction is modelled by $\vec{f} = -h\vec{v}$. Show that $x'' + \frac{h}{m}x' + \frac{k}{m}x = 0$.
5. To keep the amplitude at 10 cm, a device drives the oscillations. Calculate the average power it must supply.
6. Describe the motion of (S) if h becomes very large.

Exercise 2 — A bullet in a block — 7 pts

A block (B) of mass $M = 990$ g, fixed to a horizontal spring of stiffness $k = 100$ N/m, is at rest at O on a frictionless table. A bullet of mass $m = 10$ g, moving horizontally along the axis of the spring with speed v, gets embedded in (B). Just after the shock, at $t_0 = 0$, the system starts from O in the positive sense and oscillates with amplitude $X_m = 10$ cm.

1. Calculate ω_0 and T_0 of the oscillator (B, bullet, spring).
2. Write the time equation $x(t)$. Deduce the speed V_0 just after the shock.
3. Check V_0 using the conservation of mechanical energy of the oscillator.
4. Which physical quantity is conserved during the shock? Deduce v.
5. Show that the collision is not elastic and calculate the energy transformed into internal energy.
6. Determine the first instant when the block passes again through O and its velocity then.

Exercise 3 — Measuring g in the school lab — 6 pts

A small metal ball hangs from a light inextensible string of length ℓ; for small oscillations $T_0 = 2\pi\sqrt{\ell/g}$. The students time 20 oscillations for each length:

ℓ (m)	0.40	0.60	0.80	1.00	1.20
$20\,T_0$ (s)	25.4	31.1	35.9	40.1	44.0

1. Why must the angle of release be small?
2. Calculate T_0^2 for each length and trace $T_0^2 = f(\ell)$.
3. Determine the slope of the line and deduce the value of g.
4. A heavier ball is used with $\ell = 1.00$ m. What happens to T_0?
5. What length gives a "seconds pendulum" ($T_0 = 2$ s)? What would its period be on the Moon ($g = 1.62$ m/s^2)?